

Date: October 5, 2026

To: Editor-in-Chief, *International Journal of Numerical Methods for Heat & Fluid Flow*
Emerald Publishing

Subject: Submission of Original Research Article for Publication in *International Journal of Numerical Methods for Heat & Fluid Flow*

Dear Editor-in-Chief and Editorial Board,

We are pleased to submit our original research manuscript entitled:

**“A High-Resolution Finite-Difference Framework for Steady Continuation
and Finite-Window Unsteady Dynamics up to $Re = 100,000$ in 2D Lid-Driven
Cavity Flows”**

for consideration as a **Research Paper** in the *International Journal of Numerical Methods for Heat & Fluid Flow*.

Alignment with Journal Scope and Significance:

The *International Journal of Numerical Methods for Heat & Fluid Flow* is internationally recognized for advancing state-of-the-art computational algorithms, numerical discretization schemes, and verification benchmarks for fluid flow physics. The 2D incompressible lid-driven cavity problem represents the canonical benchmark for assessing solver accuracy, spatial dispersion, and high-Reynolds numerical stability. Established benchmark datasets, however, have historically stagnated at $Re \leq 10,000$ (Ghia et al., 1982) or $Re \leq 25,000$ (Erturk et al., 2005).

In this manuscript, we present a high-resolution finite-difference streamfunction–vorticity (ψ – ω) framework that directly advances numerical fluid dynamics across several core fronts:

1. **Exact Spectral Poisson Acceleration:** We integrate an exact Discrete Sine Transform (DST-I) Poisson solver scaling at optimal $\mathcal{O}(N^2 \log N)$ complexity, eliminating iterative convergence errors, pressure-velocity decoupling, and numerical drift.
2. **Strict Zero Numerical Viscosity ($\nu_{\text{art}} = 0$):** Spatial discretization retains pure second-order central differencing without artificial dissipation for all Reynolds numbers up to $Re = 50,000$.
3. **Directional Hybrid Regularization at $Re = 100,000$:** At extreme cell Péclet numbers ($Pe > 195$), an interior directional hybrid model regularizes driven corner singularities with explicitly mapped numerical dissipation tensors.
4. **Asymptotic Batchelor Core Verification:** Stationary continuation up to $Re = 100,000$ on uniform Cartesian meshes up to 1025×1025 (1,050,625 nodes) resolves primary and secondary eddy cascades, providing numerical evidence qualitatively consistent with Batchelor’s core homogenization theorem ($\omega_0 \approx -2.54$, $x_c \rightarrow 0.50000$).
5. **Finite-Window Unsteady Marching on 513×513 Meshes:** Physical time-accurate marching resolves supercritical Hopf bifurcations to periodic vortex shedding at $Re = 10,000$ ($St = 0.6249 \pm 0.0416$), amplitude-modulated multi-frequency dynamics at $Re = 15,000$ ($St = 0.5364 \pm 0.0278$), and finite-window spectral transitions up to $Re = 100,000$.

6. **Open Science and Benchmark Reproducibility:** Thirty (30) verified full 2D coordinate flow field datasets (23 steady continuation and 7 unsteady telemetry records) with cryptographic SHA-256 checksums are openly archived under the MIT license.

Double-Anonymous Peer Review Compliance:

In strict accordance with Emerald Publishing’s double-anonymous peer review policy, the main manuscript document has been completely anonymized, with all author names, institutional affiliations, and self-identifying statements removed. Complete author details and metadata are provided separately in the accompanying Title Page file.

Suggested Potential Reviewers:

1. **Prof. Ercan Erturk** (Bahçeşehir University, Istanbul, Turkey, Email: ercan.erturk@bau.edu.tr)
— World authority on high-Reynolds lid-driven cavity benchmarking and steady Navier–Stokes streamfunction–vorticity solutions.
2. **Prof. Olivier Botella** (Université de Lorraine, LEMTA, France, Email: olivier.botella@univ-lorraine.fr)
— Renowned specialist in high-order compact finite difference schemes and incompressible cavity benchmarks.
3. **Prof. Chang Shu** (National University of Singapore, Singapore, Email: mpeshuc@nus.edu.sg)
— Pioneer of differential quadrature, high-Reynolds FDM streamfunction–vorticity formulations, and lid-driven cavity transition analysis.
4. **Prof. Jie Shen** (Eastern Institute of Technology, China / Purdue University, USA, Email: jshen@eitech.edu.cn) — Leading expert in spectral methods, projection schemes, and Navier–Stokes stability.

Declarations:

This manuscript is an original work, has not been published previously, and is not under simultaneous consideration for publication elsewhere. All authors have read and approved the manuscript. There are no competing financial or non-financial interests to declare.

Sincerely yours,

Kartikeya Gangwar

Department of Mathematics

University of Delhi, Delhi, 110007, India

Email: kartikeyagangwar@proton.me

ORCID: <https://orcid.org/0009-0009-1973-7532>